

Experiment No: 02

Course Code: CSE-406

Course Title: Digital Image Processing Laboratory
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Submitted to-

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Experiment Name: Take an image as input and display it's negative image.

Objectives:

- **Read the Input Image:** Load the image into MATLAB.
- **Convert the Image to Negative:** Transform the pixel values to their negative counterparts.
- **Display the Original and Negative Images:** Show the images for comparison.

Source Code:

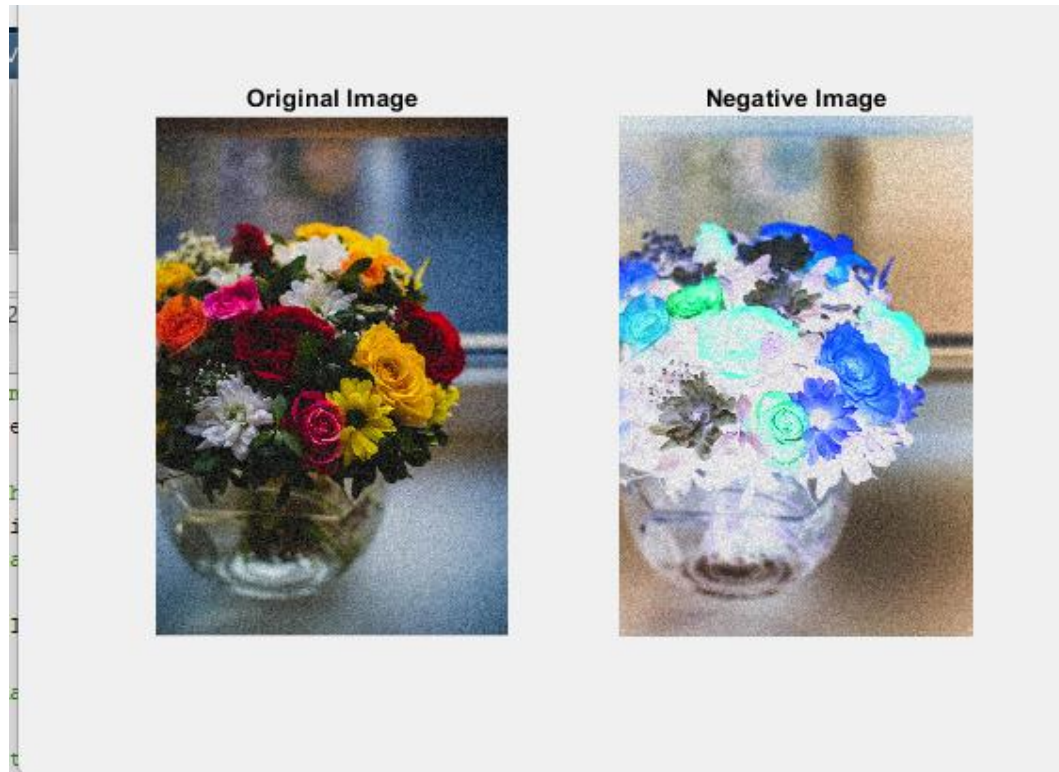
```
% Read the image
originalImage = imread('C:\Users\CSELAB-2\Desktop\GIPLab02\negativeimage1.jpg');

% Check if the image is grayscale or RGB
if size(originalImage, 3) == 1
    % Grayscale image
    maxValue = 255; % Maximum value for 8-bit images
    negativeImage = maxValue - originalImage;
else
    % RGB image
    maxValue = 255; % Maximum value for 8-bit images
    % Apply the negative transformation to each channel
    negativeImage = maxValue - originalImage;
end

% Display the original and negative images
figure;
subplot(1, 2, 1);
imshow(originalImage);
title('Original Image');

subplot(1, 2, 2);
imshow(negativeImage);
title('Negative Image');
```

Input and Output:



Experiment Name: Take an image as input and enhance it's contrast stretching image.

Objectives

1. **Read the Input Image:** Load the image into MATLAB.
2. **Define the Contrast Stretching Parameters:** Determine the minimum and maximum values for both the original and stretched image.
3. **Apply Contrast Stretching:** Transform the pixel values based on the defined parameters.
4. **Display the Original and Enhanced Images:** Show both the original and contrast-stretched images for comparison.

Source Code:

```
% Step 1: Read the RGB image
img = imread('C:\Users\CSELAB-2\Desktop\GIPLab02\contast.jpg'); % Replace
'your_image.jpg' with your image filename

% Step 2: Define the stretching parameters
min_out = 0;          % Minimum intensity of the output image
max_out = 255;        % Maximum intensity of the output image

% Initialize the stretched image
img_stretched = img;

% Process each channel separately
for c = 1:size(img, 3)
    % Get the current channel
    channel = img(:,:,c);
```

```

    % Define the input range for the current channel
    min_in = double(min(channel(:)));
    max_in = double(max(channel(:)));

    % Apply contrast stretching
    channel_double = double(channel);
    channel_stretched = ((channel_double - min_in) / (max_in - min_in)) * (max_out -
min_out) + min_out;

    % Store the result in the output image
    img_stretched(:,:,c) = uint8(channel_stretched);
end

% Step 3: Display the original and stretched images
figure; % Open a new figure window

subplot(1,2,1);
imshow(img);
title('Original RGB Image');

subplot(1,2,2);
imshow(img_stretched);
title('Contrast Stretched RGB Image');

```

Input and Output:

